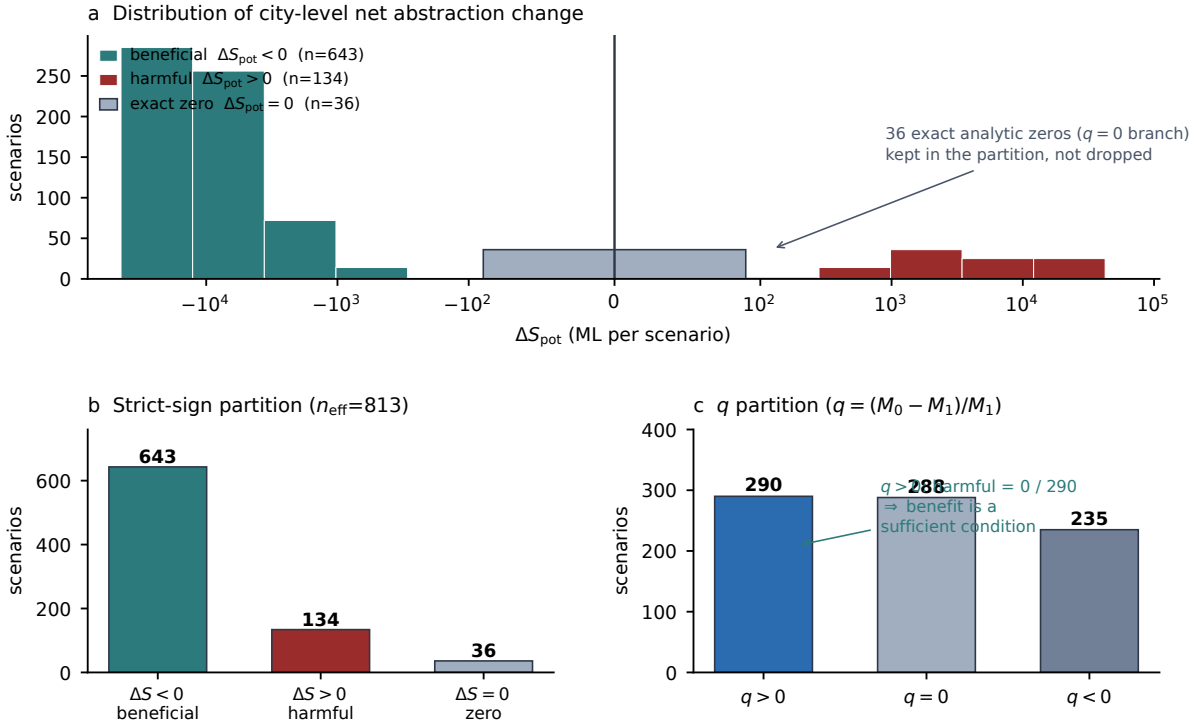

Data: A13_Reliability/supply_state_regimes.csv (a-c); column-name swap corrected per COLUMN_NAMES_CORRECTED.md; q computed as (M0-M1)/M1.
n_eff = 813 of 1052 grid rows after the dual gate (status=converged_unique AND NOT coc_floor_degenerate); 239 excluded. ZERO CONVENTION: strict sign, dS_pot == 0.0 counted separately (a tolerance band of 1e-9 gives the older 631/102 split and must NOT be used).
LIMITATION: the q>0 sufficiency check has no positive counterexample (0/290 harmful) because q>0 is algebraically tied to the same identity; it is an accounting consistency check, not an independent predictive test. City-level complete-species subset is only n_eff=47.